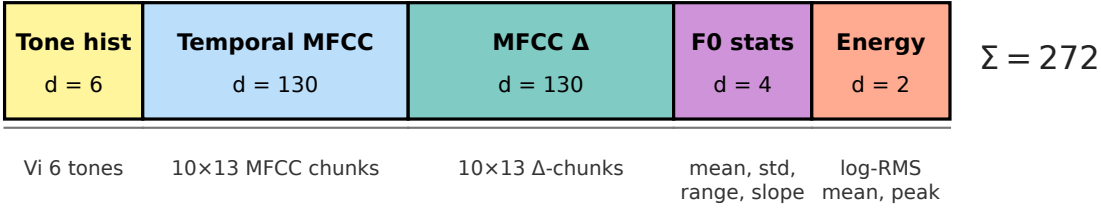
Tonal-perceptual fingerprint composition

Five concatenated, L2-normalized components → 272-d unit vector



Two-stream voting (§3.5) splits this into independent streams:
f_A = component A (6-d, tone) ⊥ f_B = components B+C+D+E (260-d, segmental)